

Detecting and Quantifying Moral Meaning Attenuation in Local Large Language Models

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Abstract—Large language models (LLMs) are increasingly used to interpret morally complex texts. A critical but underexplored failure mode is moral meaning attenuation: the weakening of moral content in LLM-generated interpretations through hedged blame, softened language, reduced agency, or weakened causal structure. Unlike hallucination or refusal, this distortion may produce no factual error, making it difficult to detect.

This study proposes a framework for detecting and quantifying this distortion by comparing a source passage S with its LLM-generated interpretation I . The method computes a composite Epistemic Fidelity Score (EFS) across four dimensions: moral explicitness, agency attribution, lexical intensity, and causal completeness. Each dimension is measured as a signed deviation between feature values extracted from S and I using standard NLP tools.

Experiments are conducted on 300 passages from dystopian literature, yielding 900 (S, I) pairs across three locally deployed models: Gemma 3 4B, Phi-4 Mini, and Qwen 2.5 14B. The framework enables automated measurement of moral-linguistic attenuation and supports cross-model comparison in a reproducible local setting.

Index Terms—large language models, moral meaning attenuation, epistemic fidelity, local models, dystopian literature

I. INTRODUCTION

Large language models (LLMs), including locally deployed models such as Gemma 3 4B, Phi-4 Mini, and Qwen 2.5 14B, are no longer merely response tools or information retrievers; they have become interpretation engines. Widely used in education, LLMs shape how researchers, educators, and scholars understand content [1], [2].

These models are trained not only to produce fluent text but also to follow instruction and safety-tuning objectives designed to reduce harmful outputs [3]. While these objectives are important, they may introduce a subtle side effect: distortion of moral meaning without altering factual correctness.

When asked to interpret morally charged texts, LLMs may hedge, generalize, or neutralize moral content [4], [5]. Unlike hallucination, where a model generates false information, or refusal, where a model declines to answer, moral meaning attenuation is a *silent* distortion. The user receives a response that appears complete and coherent, but the interpretation may weaken the source text’s moral force, agency, or causal structure [4], [6].

If left unaddressed, large-scale use of LLMs in educational and research settings risks reshaping how users interpret responsibility, causality, and historical agency [7], [8]. These

distortions may occur without factual error, making them difficult for current evaluation methods to detect.

This problem becomes particularly important in dystopian literature, a genre built around explicit structures of power, responsibility, and consequence. These texts often identify agents, assign responsibility, and preserve causal relationships between actions and outcomes. At scale, such texts shape how readers understand history, ethics, authority, and resistance [7]–[9].

This study investigates whether LLM-generated interpretations attenuate these structures: whether violence is softened into conflict, oppression is generalized into social pressure, or responsibility is obscured through neutral phrasing. Because such transformations can remain coherent, they are not captured by existing categories of hallucination or bias.

To address this gap, we shift the focus from evaluating factual correctness alone to measuring the preservation of moral-linguistic meaning. We frame moral meaning attenuation as a measurable form of semantic distortion by comparing a source passage and its corresponding LLM-generated interpretation.

Distortion is defined as divergence from the moral-linguistic fingerprint of the source across four dimensions: moral explicitness, agency attribution, lexical intensity, and causal completeness. Together, these dimensions form the basis of an Epistemic Fidelity Score (EFS), a composite signal designed to capture semantic attenuation.

In this study, we analyze 300 passages drawn from dystopian literature across Gemma 3 4B, Phi-4 Mini, and Qwen 2.5 14B, enabling comparison of moral meaning attenuation across three locally deployed models.

The rest of the paper is organized as follows. Section II reviews related work. Section III outlines the contributions. Section IV presents the EFS method. Section V describes the dataset. Section VI reports the experimental setup. Section VII describes the planned analysis, followed by validation, qualitative analysis, limitations, and conclusion.

II. RELATED WORK

LLM evaluation has evolved across multiple research directions, each targeting a distinct failure mode. This section reviews five categories and highlights gaps relevant to detecting loss of moral meaning.

Hallucination detection focuses on identifying factual errors and inconsistencies [6], [10], [11]. Although effective for

detecting incorrect outputs, these methods do not capture cases where an LLM-generated interpretation is factually accurate but attenuates the moral content of the source. In such cases, no factual inconsistency is produced, and therefore no standard hallucination signal is raised.

Previous work develops automated metrics for semantic similarity and sentiment analysis using embeddings and transformer representations [12]–[17]. These approaches measure surface-level alignment but do not model directional changes in moral structure. As a result, they may fail to distinguish between outputs that are lexically similar and outputs that preserve underlying moral relationships.

Research on bias and alignment shows that LLM outputs can reflect training-data skews and produce compliant yet distorted responses [4], [5], [7], [18]. However, these works often focus on distributional bias rather than per-instance attenuation of moral content. No framework in this area directly assigns a score to how much moral meaning a specific interpretation has lost relative to its source.

Safety and jailbreak research examines how models suppress harmful content and how safeguards can be bypassed [19]–[21]. These approaches address explicit harm but do not capture distortion in compliant outputs where meaning is altered without triggering safety mechanisms.

Recent work evaluates LLMs on narrative structure and psychological depth [9], [22], [23]. While these studies highlight structural limitations such as flattening of tension and emotional range, they do not formalize moral meaning attenuation as a measurable signal. They also do not provide a standardized metric for direct model comparison.

Category	Work	SD	MP	AA	CC	CM	AS
Hallucin-ation	Williamson [6]	○	–	–	–	–	–
	Malin [10]	–	○	–	–	–	✓
	Singh [11]	–	○	–	–	○	✓
Semantic Similarity	Rao [12]	–	○	–	–	–	✓
	Shang [13]	–	○	–	–	–	✓
	Feng [14]	–	○	–	–	–	✓
	Hany [15]	–	○	–	–	–	✓
	Horvat [16]	–	○	–	–	–	✓
	Cheng [17]	–	–	–	–	–	✓
Bias & Alignment	Dhingra [18]	–	–	–	–	–	✓
	Barnhart [4]	○	○	–	–	–	–
	Kay [5]	○	○	–	–	–	–
	Kashiramka [7]	–	○	–	–	–	–
Safety & Jailbreak	Nguyen [19]	–	–	–	–	–	✓
	Knowlton [20]	–	–	–	–	✓	✓
	Sornette [21]	○	–	–	–	–	–
Literary Eval.	Tian [22]	–	○	○	○	–	○
	Harel-Canada [23]	–	○	○	–	–	✓
	Zhou [9]	–	○	–	–	–	✓
–	Ours (EFS)	✓	✓	✓	✓	✓	✓

This table compares prior work across six challenge dimensions: silent distortion (SD), moral preservation (MP), agency attribution (AA), causal completeness (CC), cross-model comparison (CM), and automated scoring (AS).

The table highlights a consistent pattern: existing approaches address isolated aspects of the problem. Hallucination frameworks detect factual error, semantic similarity and

sentiment methods capture surface-level alignment rather than meaning preservation, bias and alignment studies show distortion but lack per-instance measurement, safety and jailbreak work focuses on explicit harm, and literary evaluation captures structural effects without formalizing them as a measurable signal.

III. CONTRIBUTIONS

We introduce a framework that uses moral-linguistic features to quantify attenuation between a source passage and an LLM-generated interpretation.

Operationalization of moral meaning attenuation. We translate epistemic harm into four measurable dimensions: moral explicitness, agency attribution, lexical intensity, and causal completeness. Each dimension is computed as a signed delta $\Delta_d = f_d(S) - f_d(I)$ and aggregated into a single Epistemic Fidelity Score (EFS).

Detection of silent distortion. The framework targets fluent and factually coherent outputs, addressing a class of distortion not captured by hallucination, bias, or toxicity metrics.

Cross-model comparison in a local setting. By applying the same pipeline to 900 (S, I) pairs across Gemma 3 4B, Phi-4 Mini, and Qwen 2.5 14B, we enable direct comparison of moral meaning attenuation across three locally deployed models.

Dystopian literature as evaluation corpus. We introduce a controlled dataset of passages with explicit moral agents, identifiable power structures, and clear causal chains.

IV. METHOD: EPISTEMIC FIDELITY SCORE

This section formalizes the environment, pipeline topology, inputs, parameters, outputs, and assumptions of the framework.

A. Environment

The system operates in an offline, post-hoc evaluation setting. Source passages and their corresponding LLM-generated interpretations are collected in advance, and the framework processes these pairs without real-time interaction with the models. The models are run locally through Ollama, supporting reproducibility and reducing dependence on external API behavior.

B. Topology

The pipeline follows a sequential five-stage architecture, as illustrated in Fig. 1:

- 1) Text ingestion: Each (S, I) pair is loaded and tokenized.
- 2) Feature extraction: Linguistic features are extracted using spaCy, VADER, and the NRC Emotion Lexicon.
- 3) Per-dimension delta computation: For each dimension d , a signed delta $\Delta_d = f_d(S) - f_d(I)$ is computed.
- 4) Score aggregation: Deltas are combined into EFS using weights w .
- 5) Model comparison: EFS distributions are computed per model for cross-model analysis.

C. Inputs

The system processes pairs (S, I) , where S is a source passage and I is its LLM-generated interpretation. The dataset consists of 900 pairs derived from 300 passages across six dystopian novels, with each passage interpreted by Gemma 3 4B, Phi-4 Mini, and Qwen 2.5 14B.

D. System Parameters

The framework includes the following parameters:

- $w = [w_1, w_2, w_3, w_4]^T$: weights assigned to the four EFS dimensions.
- λ : regularization parameter used if weights are learned through ridge regression.

When human annotations are available, weights can be learned using ridge regression. When annotations are insufficient, default weights are used and reported. The default implementation uses $w = [0.30, 0.25, 0.25, 0.20]$ for moral explicitness, agency attribution, lexical intensity, and causal completeness, respectively.

E. Outputs

For each (S, I) pair, the system produces:

- Four delta scores: $\Delta_{\text{moral}}, \Delta_{\text{agency}}, \Delta_{\text{lexical}}, \Delta_{\text{causal}}$.
- A scalar Epistemic Fidelity Score:

$$EFS = \sum_{d=1}^4 w_d \cdot \Delta_d$$

- Aggregated EFS distributions per model for cross-model comparison.

F. Assumptions

The framework rests on four assumptions:

- 1) Moral structure in source passages. Source texts contain explicit moral structure, including identifiable agents and causal chains. The framework is not intended for neutral text.
- 2) Fluency and factual coherence of interpretations. Interpretations are assumed to be factually coherent. EFS captures attenuation of moral meaning, not factual inaccuracy.
- 3) Annotation consistency. If supervised calibration is used, human annotations are assumed to provide a consistent supervision signal.
- 4) Validity of NLP tools. spaCy, VADER, and the NRC Emotion Lexicon provide approximations of the four EFS dimensions, limiting the framework to signals these tools can capture.

G. Feature Extraction and Scoring

To detect moral meaning attenuation, we model distortion as a structured divergence between a source passage S and its corresponding LLM-generated interpretation I . The proposed system follows a three-stage pipeline: feature extraction, per-dimension distortion modeling, and score aggregation.

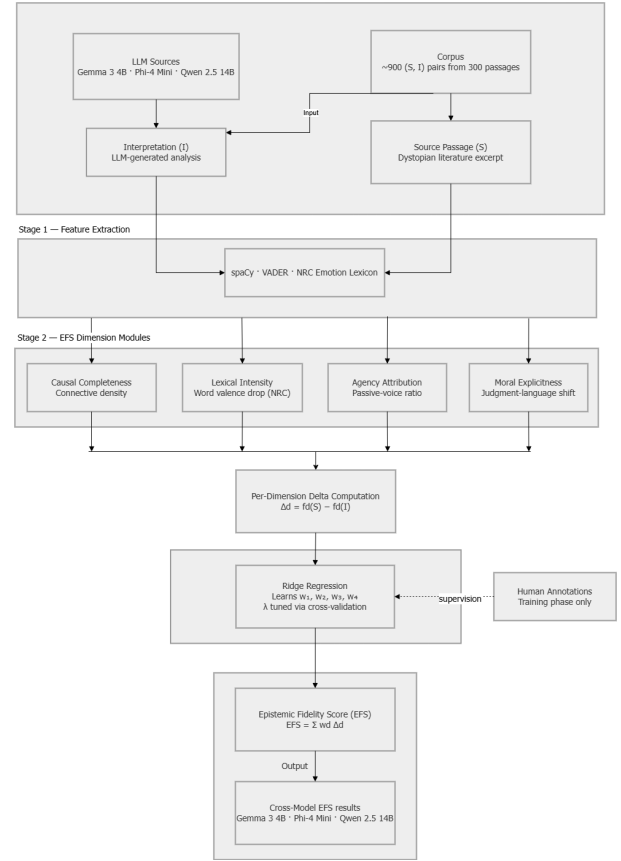


Fig. 1. End-to-end system topology.

Given a pair (S, I) , both texts are processed through a unified NLP pipeline using spaCy for tokenization, part-of-speech tagging, and dependency parsing. These representations enable the extraction of structured signals that capture how meaning is expressed.

From each text, we compute a feature vector across four dimensions:

- Moral Explicitness (f_1): Measures the presence and strength of evaluative language using polarity scoring with VADER and emotion-based intensity with the NRC Emotion Lexicon.
- Agency Attribution (f_2): Captures whether responsibility is explicitly assigned, computed using agentive constructions derived from dependency relations.
- Lexical Intensity (f_3): Quantifies the strength of descriptive language through valence and intensity shifts using lexicon-based scoring.
- Causal Completeness (f_4): Measures the preservation of cause-effect structure through connective density and dependency-based identification of causal relations.

$$F(S) = [f_1(S), f_2(S), f_3(S), f_4(S)]$$

$$F(I) = [f_1(I), f_2(I), f_3(I), f_4(I)]$$

Distortion is defined as the directional deviation between the source and interpretation along each dimension:

$$\Delta_d = f_d(S) - f_d(I), \quad d \in \{1, 2, 3, 4\}$$

Positive values indicate attenuation, where the interpretation weakens the corresponding moral-linguistic signal.

The per-dimension deviations are aggregated into a composite Epistemic Fidelity Score:

$$EFS(S, I) = \sum_{d=1}^4 w_d \cdot \Delta_d$$

where w_d denotes the contribution of each dimension. When human annotation signals are available, the weights may be learned through ridge regression:

$$w = \arg \min_w \|y - Xw\|^2 + \lambda \|w\|^2$$

where X contains the deviations Δ_d , y represents annotation signals, and λ is tuned through cross-validation.

Once EFS scores are computed, distortion is analyzed as a distribution across the dataset. High-EFS cases are inspected qualitatively to identify common forms of attenuation, such as reduced moral explicitness, weakened agency, softened lexical intensity, or loss of causal structure.

To validate the metric, each dimension can be removed in an ablation study and the resulting shifts in EFS distributions can be measured. Sensitivity analysis can also be conducted on the weights to evaluate whether model comparisons remain stable under small perturbations.

V. DATASET

The dataset consists of 300 source passages drawn from six dystopian novels: *1984*, *Fahrenheit 451*, *Animal Farm*, *We*, *The Iron Heel*, and *The Time Machine*. Each book contributes 50 passages selected for moral or political relevance using keyword-based ranking over terms associated with power, responsibility, coercion, harm, surveillance, class, and causal consequence. The observed passage length range is 83–204 words.

Book	Source Type	Passages
<i>We</i>	Project Gutenberg	50
<i>The Time Machine</i>	Project Gutenberg	50
<i>The Iron Heel</i>	Project Gutenberg	50
<i>1984</i>	Local text/EPUB-derived	50
<i>Animal Farm</i>	Local text/EPUB-derived	50
<i>Fahrenheit 451</i>	Local text/EPUB-derived	50
Total	–	300

VI. EXPERIMENTAL SETUP

Each passage is submitted to three locally deployed models, producing 900 total (S, I) pairs. The current local run has generated the model interpretations; feature extraction, EFS scoring, statistical analysis, and human validation are reported as planned analyses until the corresponding result artifacts are produced.

Model	Provider	Runtime	Model String
Gemma 3 4B	Google	Ollama/local	gemma3:4b
Phi-4 Mini	Microsoft	Ollama/local	phi4-mini
Qwen 2.5 14B	Alibaba	Ollama/local	qwen2.5:14b

A. Prompt

Each model receives the same interpretation prompt:

You are a literary analyst. Read the following passage from a dystopian novel and provide an interpretation in 3–5 sentences. Explain its meaning, themes, and significance.

The use of a fixed prompt ensures that differences in output can be compared under the same generation task. Generation is performed through Ollama’s chat interface. If model calls fail, the generation script logs the failed passage identifier and continues, allowing the run to be resumed without regenerating completed rows.

B. Planned Analyses

The evaluation focuses on model-level EFS distributions, per-dimension deltas, book-level summaries, corrected pairwise tests, effect sizes, ablation results, baseline comparisons, and qualitative inspection of high-EFS cases. Results tables will be generated only from committed CSV outputs in `data/results/`.

VII. RESULTS AND ANALYSIS

A. Overall Distortion Across Models

EFS scores were computed for all 900 (S, I) pairs. Figure 2 shows that Phi-4 Mini shifts rightward relative to zero, indicating net amplification on average, while Gemma 3 4B and Qwen 2.5 14B shift leftward, indicating net attenuation. Qwen 2.5 14B exhibits the widest spread, reflecting high per-passage variance.

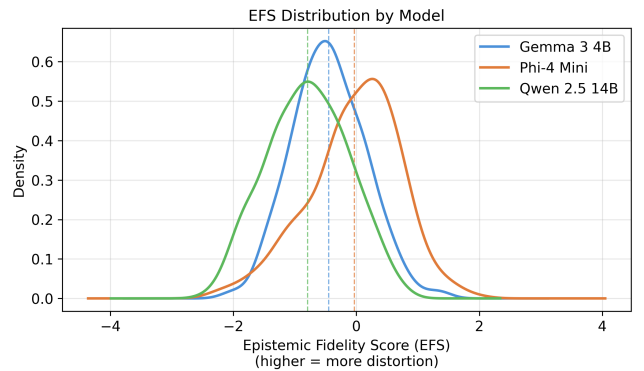


Fig. 2. EFS distributions by model. Dashed lines show per-model means.

B. Per-Dimension Analysis

Figure 3 shows mean Δ_d per model. Moral Explicitness and Agency Attribution remain near zero across all models. The dominant signal is Causal Completeness for Qwen 2.5 14B ($\approx$

−1.2), indicating systematic causal amplification. Lexical intensity is mildly negative for Gemma 3 4B and Qwen 2.5 14B. These results confirm that distortion is dimension-specific rather than uniform.

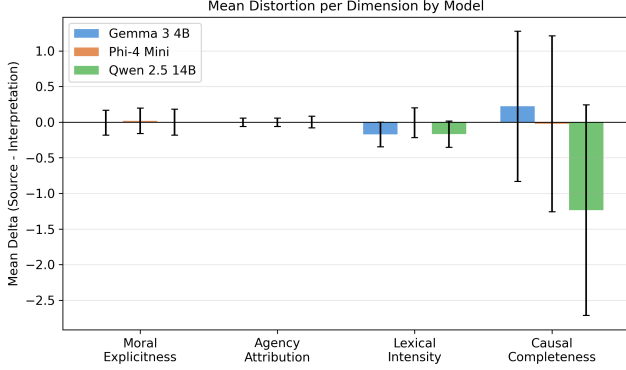


Fig. 3. Mean per-dimension deltas with standard deviation error bars.

C. Book-Level Comparison

Figure 4 reveals that attenuation patterns vary substantially by literary source. Qwen 2.5 14B produces consistently negative EFS across all six books, peaking at *The Time Machine* (≈ -1.2) and *Fahrenheit 451* (≈ -1.0). Phi-4 Mini amplifies on *The Iron Heel* and *We* while remaining near zero elsewhere. The cross-model divergence on the same source book confirms that attenuation is sensitive to both model behavior and source text structure.

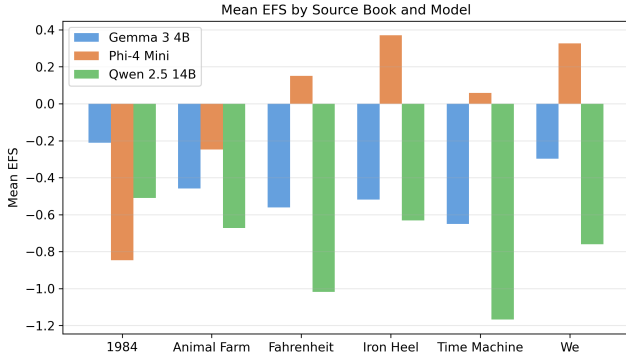


Fig. 4. Mean EFS by source book and model.

D. Ablation Study

Each dimension will be removed in turn and EFS recomputed to evaluate per-component contribution. Results will be reported in `validation_ablation.csv`.

E. Baseline, Statistical Tests, and Dimension Correlations

Figure 5 presents the Pearson correlation matrix across all 900 pairs. Inter-dimension correlations are low ($r \leq 0.33$), confirming that the four dimensions capture largely independent signals. EFS correlates most strongly with Δ_{lexical} ($r = 0.89$) and Δ_{causal} ($r = 0.59$), reflecting their higher variance in

the dataset rather than a weighting preference. Δ_{agency} is near-uncorrelated with all other dimensions ($r \leq 0.05$), indicating it captures a distinct, stable aspect of moral structure. This concentration effect motivates weight recalibration via ridge regression when annotations are available (Section VIII).

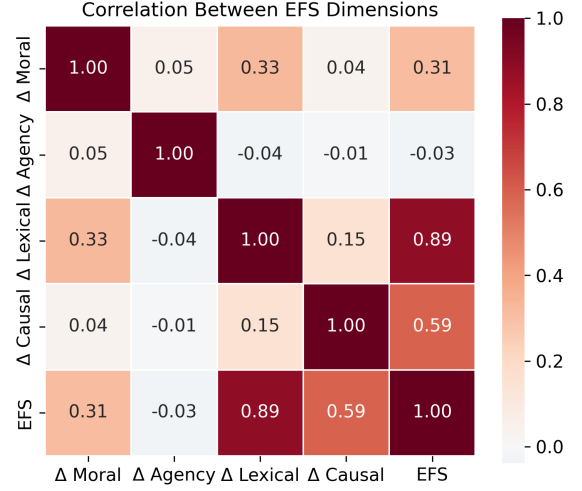


Fig. 5. Pearson correlation matrix across EFS dimensions and composite score ($n = 900$).

Table	Source Artifact
EFS summary by model	summary_statistics.csv
EFS summary by book	summary_statistics.csv
Per-dimension deltas	features_all.csv
Pairwise tests/effects	analysis_output
Ablation results	validation_ablation.csv
Baseline comparison	validation_baselines.csv
Human agreement	annotations/agreement_*.csv

VIII. HUMAN VALIDATION

Human validation will be used to test whether EFS aligns with human judgments of moral meaning attenuation. Annotators will rate shared source-interpretation pairs on a 1–5 ordinal distortion scale, where higher values indicate stronger loss of moral meaning. The minimum defensible validation set is 30 shared pairs, while the preferred target is three annotators rating the same 150 pairs.

Inter-annotator agreement will be reported before any consensus labels are used for calibration. If agreement is sufficient, consensus annotations will be used to evaluate the relationship between EFS and human judgments. When enough annotated pairs are available, ridge regression can learn dimension weights; otherwise, the default weights are retained and the limitation is reported.

IX. QUALITATIVE ANALYSIS

Qualitative analysis will use high-EFS examples from `data/results/extreme_cases.csv`. The final paper will include three to five short examples. Each example will identify what the source passage says morally, what the model interpretation changes, which EFS dimension captures

the change, and whether human annotators agreed that the interpretation was distorted.

The selected cases should include at least one agency-erasure example, at least one lexical-softening example, and at least one causal-loss example if such a case appears in the scored outputs.

X. REPRODUCIBILITY AND DATA ETHICS

All model generation uses the exact model strings `gemma3:4b`, `phi4-mini`, and `qwen2.5:14b` through Ollama, and all passages are processed through the same prompt. The code records the model key, exact model string, source passage, and generated interpretation for each row. Result tables and figures should be generated from committed CSV files so that every reported number can be traced to an output artifact.

The dataset combines public-domain Project Gutenberg texts with locally supplied or EPUB-derived texts. Copyrighted or manually supplied source texts are used only for research analysis, and examples in the paper should quote only the minimum excerpt needed for interpretation. The study evaluates textual attenuation in model outputs and does not make claims about model intent.

XI. LIMITATIONS

This study has several limitations. First, EFS approximates moral meaning through measurable linguistic features, and therefore may miss irony, implication, literary ambiguity, or broader narrative context. Second, lexicon-based and dependency-based methods are imperfect proxies for moral interpretation. Third, passage selection may be biased toward texts with explicit moral or political language.

Fourth, the correlation analysis in Section ?? reveals that EFS is most strongly driven by Δ_{lexical} and Δ_{causal} due to their higher variance across the dataset. This concentration effect means aggregate EFS values should not be interpreted as uniformly weighting all four dimensions, and weight recalibration through human annotation is recommended for future work.

Fifth, the study uses Gemma 3 4B, Phi-4 Mini, and Qwen 2.5 14B running locally through Ollama. Therefore, the findings should not be generalized to frontier API models such as GPT, Claude, or Gemini without additional experiments.

XII. CONCLUSION

This paper proposes a framework for detecting and quantifying moral meaning attenuation in LLM-generated interpretations. Unlike hallucination or refusal, this form of distortion may occur in fluent and factually coherent outputs, making it difficult to detect using standard evaluation methods.

To address this gap, we introduced the Epistemic Fidelity Score (EFS), which compares a source passage and its LLM-generated interpretation across four dimensions: moral explicitness, agency attribution, lexical intensity, and causal completeness. The framework enables automated measurement of moral-linguistic attenuation and supports cross-model comparison.

Experiments across Gemma 3 4B, Phi-4 Mini, and Qwen 2.5 14B on 300 passages from dystopian literature reveal that distortion is heterogeneous: it is dimension-specific, model-specific, and sensitive to literary source. Phi-4 Mini tends toward net amplification of moral content, while Gemma 3 4B and Qwen 2.5 14B tend toward net attenuation, with Qwen 2.5 14B exhibiting the highest variance and the strongest causal amplification signal. These patterns would not be distinguishable with a single-feature metric, supporting the value of the multi-dimensional EFS design.

Future work includes extending the framework to additional models, improving semantic feature representations, calibrating dimension weights through human annotation, and applying the method to other domains where meaning preservation is critical.

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